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**Факультет «Информатика и системы управления»
Кафедра ИУ5 «Системы обработки информации и управления»**

**Курс «Парадигмы и конструкции языков программирования»
Отчет по рубежному контролю №2
«Вариант Г, 16»**

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Листинг кода (rk2.py):

```
from operator import itemgetter

class Book:
    def __init__(self, id, title, price, shop_id):
        self.id = id
        self.title = title
        self.price = price
        self.shop_id = shop_id

class Shop:
    def __init__(self, id, name):
        self.id = id
        self.name = name

class BookShop:
    def __init__(self, shop_id, book_id):
        self.shop_id = shop_id
        self.book_id = book_id

# ----- ФУНКЦИИ ДЛЯ ТЕСТИРОВАНИЯ -----

def get_one_to_many(books, shops):
    """Связь один-ко-многим"""
    return [
        (b.title, b.price, s.name)
        for s in shops
        for b in books
        if b.shop_id == s.id
    ]

def get_shops_starting_with_a(books, shops):
    """Задание Г1"""
    one_to_many = get_one_to_many(books, shops)
    result = {}

    for s in shops:
        if s.name.startswith('A'):
            shop_books = [title for title, _, shop in one_to_many if shop == s.name]
            result[s.name] = shop_books

    return result

def get_max_price_by_shop(books, shops):
    """Задание Г2"""
    one_to_many = get_one_to_many(books, shops)
    result = []
```

```

        for s in shops:
            prices = [price for _, price, shop in one_to_many if shop == s.name]
            if prices:
                result.append((s.name, max(prices)))

        return sorted(result, key=itemgetter(1), reverse=True)

def get_books_shops_many_to_many(books, shops, books_shops):
    """Задание ГЗ"""
    temp = [
        (s.name, bs.book_id)
        for s in shops
        for bs in books_shops
        if s.id == bs.shop_id
    ]

    return sorted(
        [(b.title, b.price, shop)
         for shop, book_id in temp
         for b in books if b.id == book_id],
        key=itemgetter(2)
    )

```

Листинг кода (rk2_tests.py):

```

import unittest
from rk2 import (
    Book, Shop, BookShop,
    get_shops_starting_with_a,
    get_max_price_by_shop,
    get_books_shops_many_to_many
)

class TestBookShop(unittest.TestCase):

    def setUp(self):
        """Подготовка тестовых данных (TDD)"""
        self.shops = [
            Shop(1, 'Альфа-Книги'),
            Shop(2, 'Арбатская книга'),
            Shop(3, 'Город книг'),
        ]

        self.books = [
            Book(1, 'Алые паруса', 450, 1),
            Book(2, 'Преступление и наказание', 550, 1),
            Book(3, 'Мастер и Маргарита', 700, 3),
        ]

        self.books_shops = [

```

```

        BookShop(1, 1),
        BookShop(1, 2),
        BookShop(3, 3),
        BookShop(1, 3),
    ]

    # ----- ТЕСТ 1 -----
    def test_shops_starting_with_a(self):
        result = get_shops_starting_with_a(self.books, self.shops)
        self.assertIn('Альфа-Книги', result)
        self.assertEqual(len(result['Альфа-Книги']), 2)

    # ----- ТЕСТ 2 -----
    def test_max_price_by_shop(self):
        result = get_max_price_by_shop(self.books, self.shops)
        self.assertEqual(result[0], ('Город книг', 700))

    # ----- ТЕСТ 3 -----
    def test_many_to_many_relation(self):
        result = get_books_shops_many_to_many(
            self.books, self.shops, self.books_shops
        )
        self.assertIn(('Мастер и Маргарита', 700, 'Альфа-Книги'), result)

if __name__ == '__main__':
    unittest.main()

```

Вывод программы (rk2_tests.py):

```

(venv) koroligor@MacBook-Pro-Igor RK2 % python3 rk2_tests.py
...
-----
Ran 3 tests in 0.000s
OK

```